Classifying Bacterial Genera from 16S rRNA Sequences Using K-mer Features and XGBoost with the RDP Trainset 18 Dataset

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***Abstract*—Microbiome research depends on precise bacterial identification to elucidate microbial diversity in ecological and health-related contexts, with the 16S ribosomal RNA (rRNA) gene serving as a critical taxonomic marker. This study develops machine learning (ML) classifiers for genus-level classification using the Ribosomal Database Project (RDP) Trainset 18 dataset, comprising 21,195 FASTA-formatted 16S rRNA sequences (455–1740 base pairs, mean 1458 bp). After filtering 1,268 singleton classes, 19,927 sequences across 1,735 genera were retained. Conducted in January 2025 using Jupyter Notebook, the work addresses the limitations of alignment-based methods like BLAST by leveraging ML pattern recognition. Two modeling frameworks were developed: a Full Model encompassing 1,712 genera and a Top-100 Model targeting the 100 most frequent genera (7,822 training, 1,908 testing sequences). Preprocessing extracted a 250 bp V4 region (positions 400–650). Initial experiments with one-hot encoding for Random Forest (RF), Support Vector Machines (SVM), and Convolutional Neural Networks (CNN) yielded low accuracies (10–29%). A shift to k-mer features (4–5 bp, 8,157 binary-encoded features for Top-100; 10,986 for Full) significantly improved outcomes. XGBoost, enhanced with SMOTE for class balancing and GPU acceleration, achieved 96.91% accuracy and 97% F1-score for the Top-100 Model, while an optimized CNN reached 97.22% accuracy. The Full Model delivered 82.84% accuracy. Cross-validation ensured robustness (94.48% ± 0.58% for Top-100). Feature importance analysis identified key k-mers, enhancing taxonomic interpretability. Outperforming benchmarks like Hamed et al.'s 96.3% binary SVM, this scalable pipeline positions ML as a transformative tool for microbial ecology and health applications.**

***Index Terms*—16S rRNA, machine learning, bacterial classification, XGBoost, k-mer features, CNN, taxonomic identification, microbiome, pattern recognition, computational biology**

I. Introduction

Microbial communities underpin ecological processes like nitrogen fixation and human health functions such as immune modulation, necessitating accurate taxonomic classification. The 16S rRNA gene, with conserved and hypervariable regions (V1–V9), is the standard for bacterial taxonomy. The Ribosomal Database Project (RDP) Trainset 18 dataset, containing 21,195 sequences (mean length 1458 bp), provides a robust yet complex resource, reduced to 19,927 sequences and 1,735 genera after filtering singletons.

Traditional alignment-based tools like BLAST and QIIME, while effective for small datasets, struggle with the scale of next-generation sequencing, requiring significant computational resources and manual curation. Machine learning offers a scalable alternative, leveraging pattern recognition to classify sequences without alignment. However, challenges include non-numeric sequence data, variable lengths, and class imbalance, with dominant genera like Streptomyces overshadowing rare taxa.

Early ML attempts using one-hot encoding often lost sequence context, while k-mer feature extraction—counting short subsequences—captures local patterns efficiently. Initiated in January 2025, this study develops ML classifiers for bacterial genera, testing RF, SVM, CNN, and XGBoost on the RDP Trainset 18 dataset. A dual-model approach was adopted: a Full Model tackling 1,712 genera for scalability and a Top-100 Model focusing on 100 frequent genera for efficiency. Enhanced by SMOTE, GPU acceleration, and detailed metrics, the pipeline aims to surpass multi-class benchmarks, advancing microbial ecology, pathogen detection, and personalized medicine.

II. Literature Review

The application of 16S rRNA sequences for bacterial taxonomy originated with Woese and Fox's 1977 seminal work, which established the gene as a phylogenetic marker by analyzing conserved and variable regions to delineate prokaryotic kingdoms [1]. Woese et al.'s 1990 refinement proposed the three-domain system (Archaea, Bacteria, Eucarya), emphasizing hypervariable regions like V4 for taxonomic resolution at the genus level [6]. The advent of high-throughput sequencing expanded repositories like the RDP, necessitating computational advancements beyond manual alignment.

Yang et al.'s 2020 comprehensive review of machine learning in DNA sequence data mining identifies key challenges: non-numeric data requires transformation (e.g., one-hot encoding), and class imbalance biases models toward dominant taxa [2]. They report modest accuracies with one-hot encoding due to lost positional context and advocate for feature extraction innovations and distributed computing to scale models.

Moeckel et al.'s 2024 survey elevates k-mers as a cornerstone of genomic analysis, highlighting their computational efficiency in processing large datasets for applications like pathogen detection and cancer biomarker identification [5]. Their discussion of absent k-mers (nullomers) as diagnostic tools suggests future interpretability avenues, though 16S rRNA specificity is limited.

Hamed et al.'s 2023 study integrated pattern-matching with ML, achieving 96.3% accuracy using a linear SVM for binary DNA classification [3]. Their preprocessing pipeline, converting FASTA to CSV and extracting features, mirrors our approach, while their focus on execution time (e.g., 0.0012 seconds for 25-length patterns) informs our efficiency goals.

Caporaso et al.'s 2010 QIIME pipeline remains a standard for 16S rRNA analysis, integrating alignment and clustering but facing scalability challenges with large datasets [4]. Jackman et al.'s 2017 ABySS 2.0 advanced genomic assembly with Bloom filters, offering insights into efficient k-mer handling [7]. Chen and Guestrin's 2016 XGBoost framework formalized gradient boosting, excelling in high-dimensional, multi-class tasks, making it ideal for our 1,712-class challenge [8]. Edgar's 2018 USEARCH introduced alignment-free sequence analysis, inspiring our k-mer shift [9].

Recent applications of convolutional neural networks (CNNs) in sequence classification, as noted by Yang et al., suggest potential for 16S rRNA tasks, motivating our CNN experiments. This study synthesizes Woese's biological foundation, Yang's ML challenges, Moeckel's k-mer advocacy, Hamed's pattern-matching, and technical innovations from Caporaso, Jackman, Chen, and Edgar, applying k-mers, XGBoost, and CNN to a complex multi-class 16S rRNA classification problem.

III. Methodology

A. Dataset and Preprocessing

The RDP Trainset 18 dataset included 21,195 16S rRNA sequences (455–1740 bp, mean 1458 bp), annotated with hierarchical taxonomic labels (e.g., "Root;Bacteria;Actinobacteria;Streptomyces"). Singletons (1,268 classes with one sample) were filtered, yielding 19,927 sequences across 1,735 genera to ensure robust train-test splits. An 80:20 split produced 15,941 training and 3,986 testing sequences, using a random seed (42) for reproducibility.

B. Feature Extraction

Initial experiments employed one-hot encoding, representing each nucleotide (A, T, C, G) as a 4-dimensional vector (e.g., A=[1,0,0,0]), flattening the 250 bp V4 region into 1000-dimensional vectors for CNN inputs. Due to poor performance, we pivoted to k-mer features using scikit-learn's CountVectorizer. For the Top-100 Model, 4–5 bp k-mers (e.g., "ATGC," "TGCA") were extracted with binary encoding, producing 8,157 features to balance specificity and dimensionality. The Full Model used 4–5 bp k-mers with count-based encoding, yielding 10,986 features. K-mers captured local sequence patterns, forming sparse matrices that preserved taxonomic context. Vectorizers and feature matrices were saved as checkpoints for efficiency.

C. Model Development

Four models were tested: RF (100 trees), SVM (linear kernel), CNN, and XGBoost, with XGBoost and CNN emerging as the most effective. The CNN, applied to the Top-100 Model, featured a three-layer architecture with 128, 128, and 64 filters (kernel sizes 8, 8, 3), batch normalization, max pooling, and dropout (0.2–0.5) to prevent overfitting. It was trained for 30 epochs using the Adam optimizer (learning rate 0.001) on one-hot encoded inputs, with early stopping (patience=5) to optimize performance.

XGBoost was used for both models. For the Top-100 Model, it employed 200 trees, a maximum depth of 10, a learning rate of 0.05, subsample ratio of 0.8, column sampling of 0.8, and minimum child weight of 3, minimizing multi-class log loss with L1/L2 regularization. SMOTE oversampling expanded the training set to 84,100 samples to address class imbalance. The Full Model used 100 trees, a maximum depth of 8, and class weights computed via compute_class_weight. Five-fold cross-validation with 3–4 bp k-mers (740 features, 50 trees) validated the Top-100 Model's robustness. GPU acceleration (tree_method='gpu_hist') was leveraged for both XGBoost models.

D. Implementation Details

BioPython's SeqIO module parsed FASTA files, trimming sequences to the 250 bp V4 region (positions 400–650) to ensure taxonomic relevance and uniformity across variable-length inputs. Taxonomic labels were extracted from sequence descriptions (e.g., second token) and reindexed to integers: 0–1,711 for the Full Model and 0–99 for the Top-100 Model, enabling efficient multi-class classification.

K-mer features were generated using CountVectorizer, producing sparse matrices of 4–5 bp k-mer counts for the Full Model and binary-encoded k-mers for the Top-100 Model. XGBoost constructed decision trees, splitting on k-mer counts (e.g., "ATGC" ¿ 5), and optimized multi-class log loss via gradient descent with second-order derivatives, accelerated by the RTX 4080 GPU. SMOTE balanced rare classes in the Top-100 Model, enhancing performance for underrepresented genera.

The CNN processed one-hot encoded sequences as 4-channel matrices (500 bp × 4 bases, padded or truncated), with convolutional layers detecting sequence motifs, followed by max pooling and dropout. Performance metrics, including accuracy, top-5 accuracy, precision, recall, and F1-score, were computed, and feature importance analysis identified key k-mers (e.g., "CGTA") for taxonomic differentiation. Results were saved as checkpoints, and performance summaries were tabulated for analysis.

IV. Results

Initial experiments with one-hot encoding produced poor results. RF achieved 0.74% accuracy and 2.02% top-5 ac-

TABLE I
PERFORMANCE METRICS FOR TOP-100 AND FULL MODELS

Model	Dataset	Accuracy %	F1 Score %	Training Time (s)
XGBoost	Top-100	96.91	97	640
CNN	Top-100	97.22	97	419
XGBoost	Full	82.84	-	503

TABLE II
COMPARISON WITH LITERATURE BENCHMARKS

Study	Classes	Accuracy (%)	Notes
XGBoost (Top-100)	100	96.91	SMOTE, GPU
CNN (Top-100)	100	97.22	Optimized
XGBoost (Full)	1,712	82.84	Scalable
Hamed [3]	2	96.3	Binary only
QIIME [4]	Multi	-	Limited scale

curacy on 942 test samples. SVM yielded 10.83% accuracy and 23.14% top-5 accuracy, while an early CNN configuration reached 10.83% accuracy and 28.56% top-5 accuracy. These models were biased toward dominant genera (e.g., Streptomyces, 510 samples) due to lost sequence context.

Transitioning to k-mer features significantly improved performance, as summarized in Table I. For the Top-100 Model, XGBoost with 4–5 bp k-mers (8,157 binary-encoded features) and SMOTE (84,100 training samples) achieved 96.91% accuracy, 99.42% top-5 accuracy, and a 97% weighted F1-score across 1,908 test samples. Five-fold cross-validation confirmed robustness at 94.48% (±0.58%) in 71 seconds. The optimized CNN, using one-hot encoding, reached 97.22% accuracy, 99.53% top-5 accuracy, and a 97% F1-score, slightly outperforming XGBoost. Notable performance was observed for rare classes (e.g., Class 53: F1=0.67, 2 samples), reflecting SMOTE's effectiveness. The Full Model, using XGBoost with 4–5 bp k-mers (10,986 features) and class weights, delivered 82.84% accuracy and 91.57% top-5 accuracy on 3,940 test samples, though detailed per-class metrics are pending.

Feature importance analysis revealed key k-mers (e.g., "CGTA," "ATGC") driving taxonomic predictions, suggesting links to V4 region motifs. Class distribution analysis highlighted imbalance (e.g., Class 24: 211 samples vs. Class 53: 2 samples), mitigated by SMOTE in the Top-100 Model. The F1-score distribution (mean 0.96) indicated balanced performance across classes.

Table II compares our results to literature benchmarks. The Top-100 Model's 96.91–97.22% accuracy surpasses Hamed et al.'s 96.3% binary SVM and QIIME's alignment-based approach. The Full Model's 82.84% accuracy demonstrates scalability for 1,712 classes, a significant advancement over traditional methods.

V. DISCUSSION

The pipeline's success stems from k-mer features, which capture local sequence patterns critical for taxonomic differentiation. Table I illustrates the Top-100 Model's high performance, with XGBoost (96.91% accuracy, 97% F1-score) and CNN (97.22% accuracy) outperforming earlier one-hot encoding attempts (10–29% accuracy). The CNN's slight edge and faster training (419s vs. 640s) highlight its potential, contradicting initial low results due to a less optimized architecture.

XGBoost's robustness, enhanced by SMOTE's expansion to 84,100 training samples and GPU acceleration, excels in handling imbalanced, high-dimensional data. Binary k-mer encoding for the Top-100 Model reduced dimensionality (8,157 features) compared to prior 3–6 bp k-mers (10,036 features), maintaining high accuracy. Feature importance analysis identified k-mers like "CGTA" as taxonomically informative, offering a step toward biological interpretability, though mapping to V4 motifs requires further validation.

Table II benchmarks our results against Hamed et al.'s 96.3% binary SVM and QIIME, demonstrating superior multi-class performance. The Full Model's 82.84% accuracy reflects the complexity of 1,712 classes, surpassing alignment-based methods in scalability. SMOTE improved rare class performance (e.g., Class 53: F1=0.67), but detailed metrics for the Full Model are needed. The 250 bp V4 focus may miss longer-range signals, and computational costs, though mitigated by GPU acceleration, suggest opportunities for feature selection.

Future work will apply SMOTE to the Full Model, explore 6–7 bp k-mers, develop hybrid CNN-XGBoost models, and map k-mers to taxonomic motifs for enhanced biological insights.

VI. CONCLUSION

This study demonstrates the efficacy of k-mer features, XGBoost, and CNN for 16S rRNA sequence classification, achieving 96.91–97.22% accuracy for the Top-100 Model and 82.84% for the Full Model. SMOTE, GPU acceleration, and binary k-mer encoding address class imbalance and scalability, while feature importance analysis enhances taxonomic interpretability. Outperforming binary and alignment-based benchmarks, the pipeline offers a scalable solution for microbiome research, with applications in pathogen detection and personalized medicine. Future efforts will optimize the Full Model, explore hierarchical classification, and deepen biological insights through k-mer mapping.

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